



Spatial assessment of the availability of healthcare facilities at district level in Balochistan using GIS: identifying gaps and way forward

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Abstract

The availability of healthcare facilities equipped with allied health services is crucial for achieving better health. Balochistan, the largest province of Pakistan by area, faces severe problems in ensuring the availability of healthcare facilities at the district level. This study aims to assess the geospatial analysis of the distribution and availability of healthcare facilities at the district level across Balochistan province. Data from global and national open sources were used to generate maps showing the location of healthcare facilities at the district level across the province. The data on healthcare facilities were collected from the Health Department of the Government of Balochistan. The geospatial analysis was carried out using ArcGIS 10.8.2 software. Additionally, the health care services were overlaid with the district's population density to measure availability. The findings revealed the categorization of districts as low, medium and high based on the availability of healthcare facilities present in each district concerning population. The findings revealed that the number of hospitals in the districts of Balochistan varies i.e., availability is low in 13 districts, medium in 12 districts and high in 7 districts.

Keywords Geospatial analysis · Geographical information system (GIS) · Health care facilities · Balochistan · Pakistan

1 Introduction

The availability of adequate infrastructure supported by necessary diagnostic equipment, trained staff and drugs is imperative for the effective delivery of healthcare services [1]. Developing countries are prone to challenges and problems in the healthcare framework, so the availability of healthcare services is considered a problem even in developed countries. Several factors affect access to healthcare

services, such as the geographical location of health facilities, conditions of paths/roads leading to health facilities, the economic condition of healthcare-seeking individuals/groups, cultural barriers, poor governance, inadequately trained healthcare staff, and inequity in health workforce distribution. Penchansky and Thomas have proposed their model of access to health care describing the concept of fit between community healthcare needs and the healthcare system's ability to meet their needs [2]. Their proposed model is known as a 5-As model, which includes availability, accessibility, affordability, acceptability and accommodation of health care. Panezai [3] examined the refinement of the Thomas and Penchansky model by Frenk 1992, who posited that availability is the existence of productivity or the capacity of healthcare resources to generate the desired services.

The global literature has identified the importance of adequate healthcare facilities in improving healthcare outcomes. Adequate availability of public health facilities leads to better utilization of MCH services at the district level in India [4]. Research has found that a weak administrative system, poor availability of child healthcare services and limited readiness of healthcare facilities are linked with a greater likelihood of under-five mortalities in Bangladesh

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[5]. Research studies also emphasize improving the quality of health services at healthcare facilities to ensure equity in access to healthcare facilities in Nigeria [1]. A baseline health facility survey in 10 low-income and middle-income countries found that excluding hospitals, the availability of primary care tended to be lower, which needs to be improved to get universal health coverage [6]. Similarly, in Pakistan, the public healthcare system faces several challenges, such as a shortage of resources, inadequacy and lack of a trained health workforce, and access to healthcare services [3, 7].

Geographical Information System (GIS) based mapping and spatial analysis are essential in policy and strategy-making for the health sector in a region. The geospatial analysis of health facilities provides insights to link people with locations through defining optimized locations for healthcare facility establishment. To tackle the lack of access to healthcare services, there is a need to identify all the healthcare facilities across a region by creating maps with coordinates of all healthcare facilities. Map generation of all public healthcare facilities in an area highlights the concentration of healthcare facilities, indicating the number of healthcare units available in different districts.

Several studies have been conducted in Pakistan to identify the factors contributing to an uneven public health care system [8]. has pointed out aspects of the poor healthcare provision system and discussed the ineffectiveness of healthcare services in Pakistan due to the weak governance in the public healthcare system. Chotchoungchatchai et al. (2020) [9] discussed the absence and uneven delivery of the health workforce, resulting in a lack of access to healthcare services in the country. Similarly, Ahmed et al. (2024) [10] identified resource constraints, insufficient investment, lack of oversight, inadequate training of healthcare providers, and inadequate availability of essential medicines and equipment are found to be the key challenges. Research has also found that Pakistan is struggling to provide essential primary health care services, particularly in the rural and underserved areas [11].

In Balochistan, the public health care system has limited resources, poor governance, staff absenteeism, lack of health policy monitoring, and a shortage of equipment and medicines (Panzai et al., 2017) [3]. Panzai (2012) [12] has found significant inter-district variation in health care services and development in Balochistan. Similarly, Ali & Panzai (2021) [13] and Saleh et al. (2021) [14] have assessed the quality of primary health care (PHC) services at the basic health units (BHUs) level in Quetta, Balochistan's provincial capital. Similarly, Panzai et al. (2017, 2019, 2020b) [3, 15, 16] have assessed the quality of PHC services at the BHU level in rural areas of the Pishin district. Moreover, Hameed et al. (2021) [17] have assessed the maternity and child health and family planning services at the BHUs level

in Panjgur district, Balochistan. Compared to the national routine immunization coverage of 66%, Balochistan has the lowest of 29% [18]. However, limited attention has been paid to assessing Balochistan's spatial distribution of healthcare facilities. Balochistan is, although the largest province of Pakistan by area, the least developed province [18], determining the spatial distribution of public health care facilities is highly significant. The detailed visual representation of health care services, on the one hand, would provide insight into the availability of health care facilities, while on the other hand, it would enable the government and planning and development department to choose optimal locations for establishing new health care facilities. Thus, this study aims to assess the geospatial analysis of the distribution and availability of healthcare facilities at the district level across Balochistan province. Availability refers to the location and capacity of services to meet the healthcare needs of the community [19].

2 Materials and methods

2.1 Study area

Balochistan is Pakistan's largest province by area, with an area of 347,190 km², forming 43.6% of the country's total land area. The majority of the population of the province is rural [20]. It is located in the country's southwestern part, with coordinates covering 22.0 N to 32.0 N latitudes and 66.0E to 70.0E longitudes (Fig. 1). Quetta is the provincial capital of Balochistan, with a majority of Pashtun tribes and minorities of Baloch, Hazara, and Punjabi [21]. According to the 7th Population and Housing Census-2023, the population of Balochistan is 14.89 million, with a low population density of 42.90 persons per/km² [22]. The province has an arid and semi-arid climate with annual precipitation from 200 to 350 mm. About 80% of the province area is inter-mountainous, whereas flood and coastal plains cover the remaining 20%. The significant mountain ranges in the province are Suleman, Toba-Kakar, Kirthar, Chagai, Raskoh, Central Brahui, central Makran, and the Makran coastal range. The Baloch and Pashtuns are the major tribes in Balochistan; the Baloch tribes make up the majority in the province's south and east, while the Pashtuns tribes make up the majority in the north. The main economic sectors of the province are livestock, agriculture, fisheries, and mining minerals such as coal, chromite, copper, etc. The annual urban growth in the province is 4.5% [21]. People in the province are moving from rural to urban areas to get easy access to basic facilities such as health care, education, etc. The people residing in the province's capital have better healthcare services than those living in other urban and

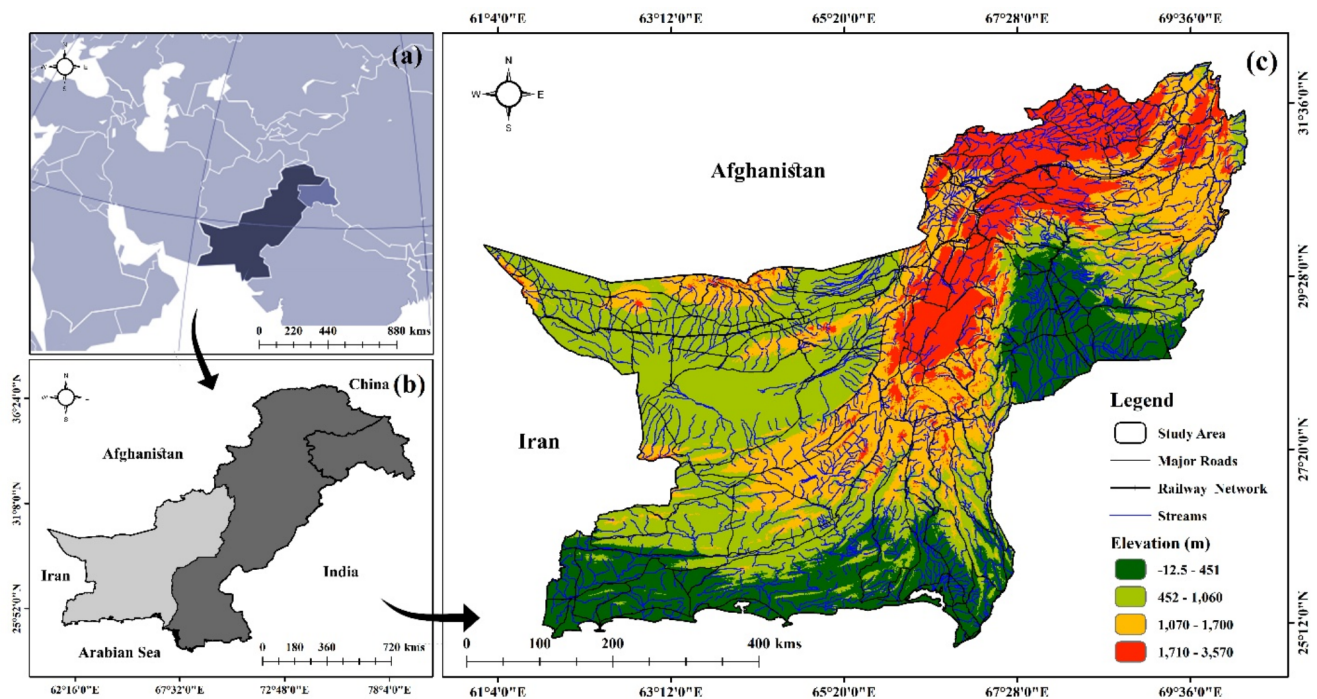


Fig. 1 (a) Pakistan's association with neighboring countries, (b) Location of the study area in Pakistan, (c) Study area map with major roads, railway network, streams, and elevation

rural areas due to the presence of private-sector hospitals and clinics.

2.2 Health care delivery system

Healthcare provision in Pakistan is based on a three-tier system, including public, private, and non-governmental sectors [23]. The public health sector provides services to 22% of the overall population, and the private health sector covers the rest of 78% of the population in the country [24]. To ensure the effective delivery of health care, the responsibility for providing health care services was transferred to the provincial government under the constitution of Pakistan through the devolution of services in 2010 [3]. Along with this, some other efforts have been made by the government to improve the healthcare delivery system, such as participating in the Millennium Development Goals (MDGs) and Sustainable Development Goals (SDGs) by expanding the Primary Health Care (PHC) facilities such as Basic Health Units (BHUs), Rural Health Centers (RHCs) and Tehsil Headquarter (THQ) hospitals, District Headquarter (DHQ) hospitals and Tertiary Health Care (THC) hospitals i.e., teaching hospitals [25]. The PHC facilities are supposed to be located in rural areas within a radius of 5 km of the population. The provincial governments also run several vertical health programs such as the Expanded Program on Immunization (EPI), AIDS Control Program, National Tuberculosis Control Program, Malaria Control Program, National

Program for Prevention of Blindness, National Program for Family Planning and Primary Health Care, National Maternal, Neonatal, and Child Health Program (MNCH), National Nutrition Program, and National Program for Prevention and Control of Hepatitis [12].

2.2.1 Hospitals

Hospitals play a major role in providing public healthcare facilities containing equipment for diagnosis and other medical purposes. The availability of hospitals and trained staff ensures a high level of health care, and analyzing a hospital's geographic aspect is pivotal for healthcare availability. Hospitals are located in both urban and rural areas. For instance, the District Headquarters (DHQ) Hospital is in urban areas, and Tehsil Headquarters (THQ) Hospital is in rural areas and towns.

2.2.2 Rural health centers (RHCs)

To overcome the inequities in the availability of healthcare in rural areas, Rural health centres (RHCs) are crucial healthcare facilities. The geographical isolation of the population, especially in the case of Balochistan, poor health infrastructure, and limited availability of health care staff are the main hurdles faced by rural populations seeking health care. RHCs play a pivotal role in enhancing the availability of healthcare services for remote communities. RHCs provide

health care services, including preventive care, mother and child health care, basic diagnostics/tests, and treatment for common illnesses. RHCs also reduce the burden of patients on hospitals and provide health care for elderly patients. At the tehsil level in rural areas of the district [12].

2.2.3 Basic health units (BHUs)

Basic Health Units (BHUs) are an important primary health-care provision component. BHUs are established at the most basic level of health infrastructure and are located in both urban and rural areas [12]. Basic healthcare such as immunization, vaccinations, treatment for common illnesses and basic diagnostics is provided by the BHUs. These basic-level health facilities provide a base for reporting and addressing health care at the grassroots level and guide patients who require a higher level of medical care. Awareness programs for sanitation, disease prevention, and healthy practices are carried out at BHUs as they are engaged with the population at a community level. BHUs are crucial in enhancing timely access to medical care. The cost-effectiveness of the treatments provided at BHUs makes health care more financially accessible to those less fortunate. It spares them the expense of travelling to far-flung urban health centres for medical care. BHUs actively interact with the local population and accommodate the cultural preferences of the area.

2.2.4 Civil dispensary

The Civil Dispensary (CD) provides basic health care to the public and is headed by a physician or dispenser. These are located mostly in urban areas of the district. However, in some cases, these may operate in rural areas as well [12].

2.2.5 Mobile dispensary

MD provides health care to remote areas without proper health care establishments. By moving to those regions and

catering to the health needs of isolated populations, extending free medical consultations and basic medical treatments, and, in some cases, providing vaccination services. These MDs are led by lady health visitors (LHVs), community midwives (CMWs) and, in some cases, medical doctors.

2.2.6 Maternity and child health

Maternity and Child Health (MCH) Centers are vital in delivering health care to mothers and newborn babies and are specifically designed for mother and child health. These are located both in urban and rural areas throughout the country.

2.3 Operational definition of availability

According to Penchansky and Thomas's model, availability is defined as the relationship between the number of health-care resources and services to the healthcare needs [3]. Similarly, availability is also defined as the location and capacity of services to meet consumer and community demand [19]. For the current study, the availability of healthcare facilities is defined as the number of healthcare facilities such as hospitals, RHCs, BHUs, CD, MD, and MCH centres located in the districts of Balochistan province.

2.4 Data collection, analysis and processing

2.4.1 Dataset sources

Several data sources are used in the current study for the geospatial analysis of public healthcare facilities in Balochistan. Shuttle Radar Topography Mission (SRTM) digital elevation model (DEM) with a resolution of 30 m was acquired from the United States Geological Survey (USGS) website (Table 1) to extract the elevation and streams of the study area. The data on Pakistan administrative boundaries, as a shapefile, was collected from the Humanitarian Data Exchange (HDX), which is an open-source archive of humanitarian data. District boundaries of Balochistan were extracted from the collected data. The population density data of the census 2017 was collected from the Pakistan Bureau of Statistics (PBS) [26]. Data for the population of each district was extracted from the collected data and added to the shapefile. Data on the health care facilities were collected from the health department of the government of Balochistan. The detailed description of online dataset sources used for the current study is shown in Table 1, and the methodological framework used for this study is presented in (Fig. 2).

Table 1 A detailed description of the datasets used in the present study is provided

S.No.	Data	Year	Data Sources
1	SRTM DEM (30 m)	2024	USGS https://earthexplorer.usgs.gov/
2	Administrative boundaries	2024	Humanitarian Data Exchange (HDX) https://data.humdata.org/
3	Population density	2017	Pakistan Bureau of Statistics (PBS) https://www.pbs.gov.pk/content/brief-census-2017
4	Health	2024	Provincial Health Directorate (GoB) http://www.dghs.gob.pk/

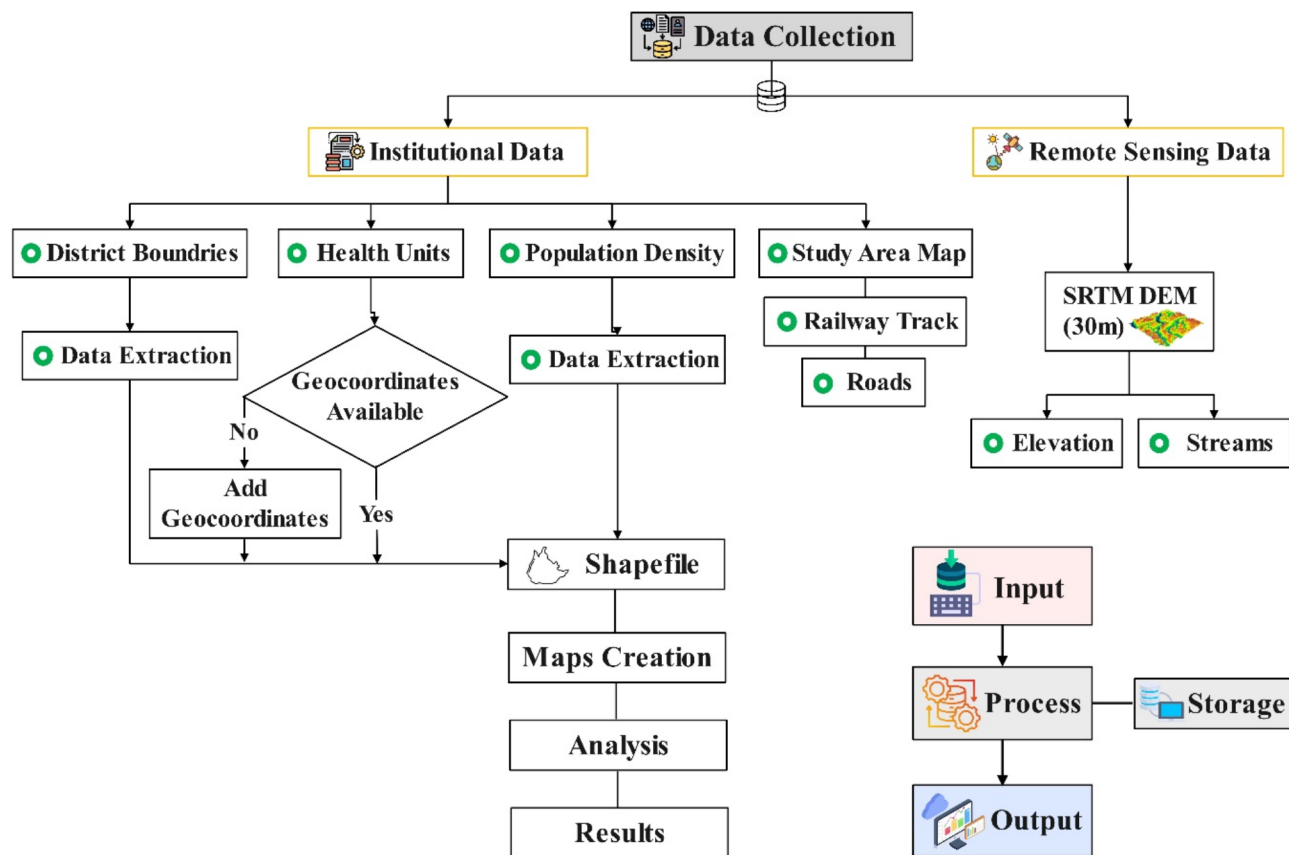


Fig. 2 Methodological framework used for the present study

2.4.2 Data analysis and processing

The second step in performing the geospatial analysis of healthcare units across the Balochistan province was analyzing the acquired datasets after data acquisition. During this process, the data was extracted, cleaned, and corrected. The data on healthcare facilities were summarized based on their occurrence and available geographic information. The data of public health care facilities were processed for statistical analysis and mapped for visualization.

Data on healthcare facilities were collected from the health department of the Government of Balochistan. Relevant data, including name, type, and location of healthcare facilities, were extracted from the collected data. As per data from the Director General of Health, Balochistan, out of the 1515 healthcare facilities, 725 lacked geo-coordinates. Geo-coordinates of 248 healthcare facilities were added by searching and finding the coordinates on Google Maps, while the geo-coordinates of 477 healthcare facilities were not found. The health units lacking geo-coordinates were added for statistical calculation. However, they were not included in the mapping process. The healthcare facilities being extracted were categorized into Hospitals, Rural Health Centers (RHCs), Basic Health Units (BHUs), Civil

Dispensaries (CD), Mobile Dispensaries (MD), Maternal and Child Health (MCH) Centers and. The MD, CD, and MCH were merged into a single category.

In this study, ArcGIS 10.8 was utilized to work with geographical information and perform geospatial analysis. It lays out geographic statistics in a visualized manner through layers. District administrative boundaries were extracted as a shapefile (the file format used in ArcGIS), and the population of each district was added to the attribute table of the shapefile, designating each district area with its population. Extracted and corrected data of hospitals, RHCs, BHUs, CD, MD, and MCH centres were aligned separately in Comma Separated Values (CSV) files. The CSV file was then imported into ArcGIS 10.8 using the import tool and was added as a layer. The collected, extracted and corrected data of district boundaries, population, and health care units were stored as attributes in the shapefile.

2.5 Mapping

ArcGIS 10.8 was used to generate maps representing population density and availability of health care services across all districts of Balochistan. The health facilities (i.e., hospitals, BHUs, RHCs, CD, MD, and MCH Centers) in all

districts of the province were mapped thematically. These thematic maps were generated by partitioning results into a three-point scale: “low” representing less, “medium” showing average, and “high” representing the higher number of health facilities per population. For classification purposes, the number of healthcare facilities in a specific district is compared with other healthcare facilities. Health facilities per population in each district were calculated, and a range was specified for defining the low, medium, and high classes in maps. The calculation was carried out for each type of healthcare facility (i.e., Hospitals, BHUs, RHCs, CD, MD, and MCH Centers).

3 Results

The population of Balochistan is dispersed along the vast landscape. Mountains, hills, plateaus, plains, and deserts cover the land of Balochistan. Population density is concentrated in the capital of the province, Quetta city. Population per district is represented in Fig. 3.

3.1 Spatial distribution of health care facilities

3.1.1 Hospitals

The availability of healthcare facilities in Balochistan varies from district to district based on population density. Districts in the province were categorized into low, medium, and high, representing the number of healthcare facilities. As shown in Fig. 4, the population is dispersed. Hospitals in these districts are also inadequate in numbers to fulfil the population's health needs. The districts of Nasirabad and Sohbatpur have hospitals that are represented in all districts of Balochistan, highlighting the regions that require the establishment of new hospitals to meet the needs of the population. Districts in the northeastern province, including Killa Abdullah, Pishin, Ziarat, Harnai, Kohlu, Zhob, Sherani, and Musakhel, require more hospitals. Khuzdar, being the central district, also has an insufficient number of hospitals. Districts of Panjgur and Kech have large land areas. In contrast, comparatively lesser land areas lie in the south-eastern part of Balochistan, and there is a lack of sufficient hospitals for population health coverage.

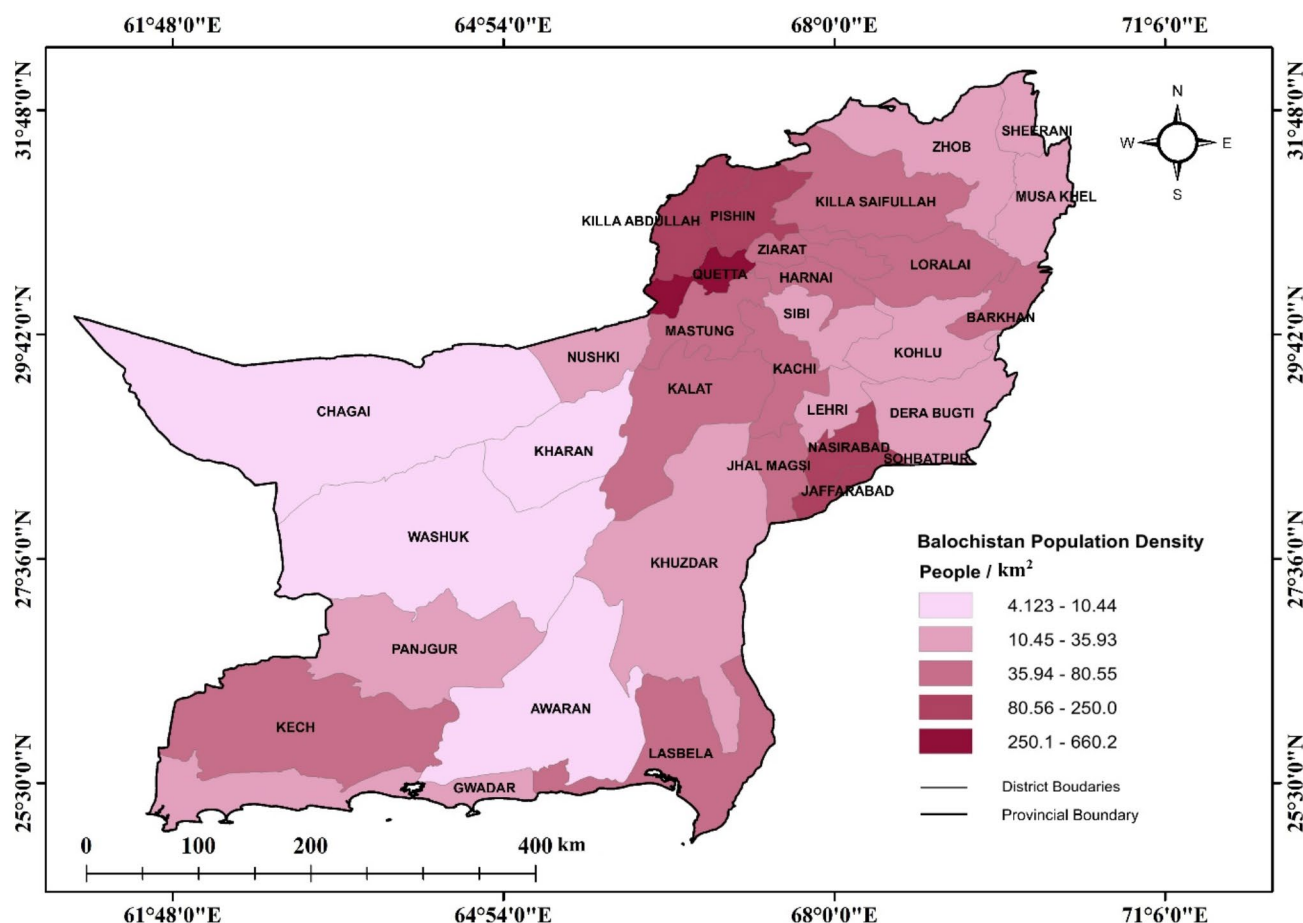


Fig. 3 Population density at the district level in Balochistan province

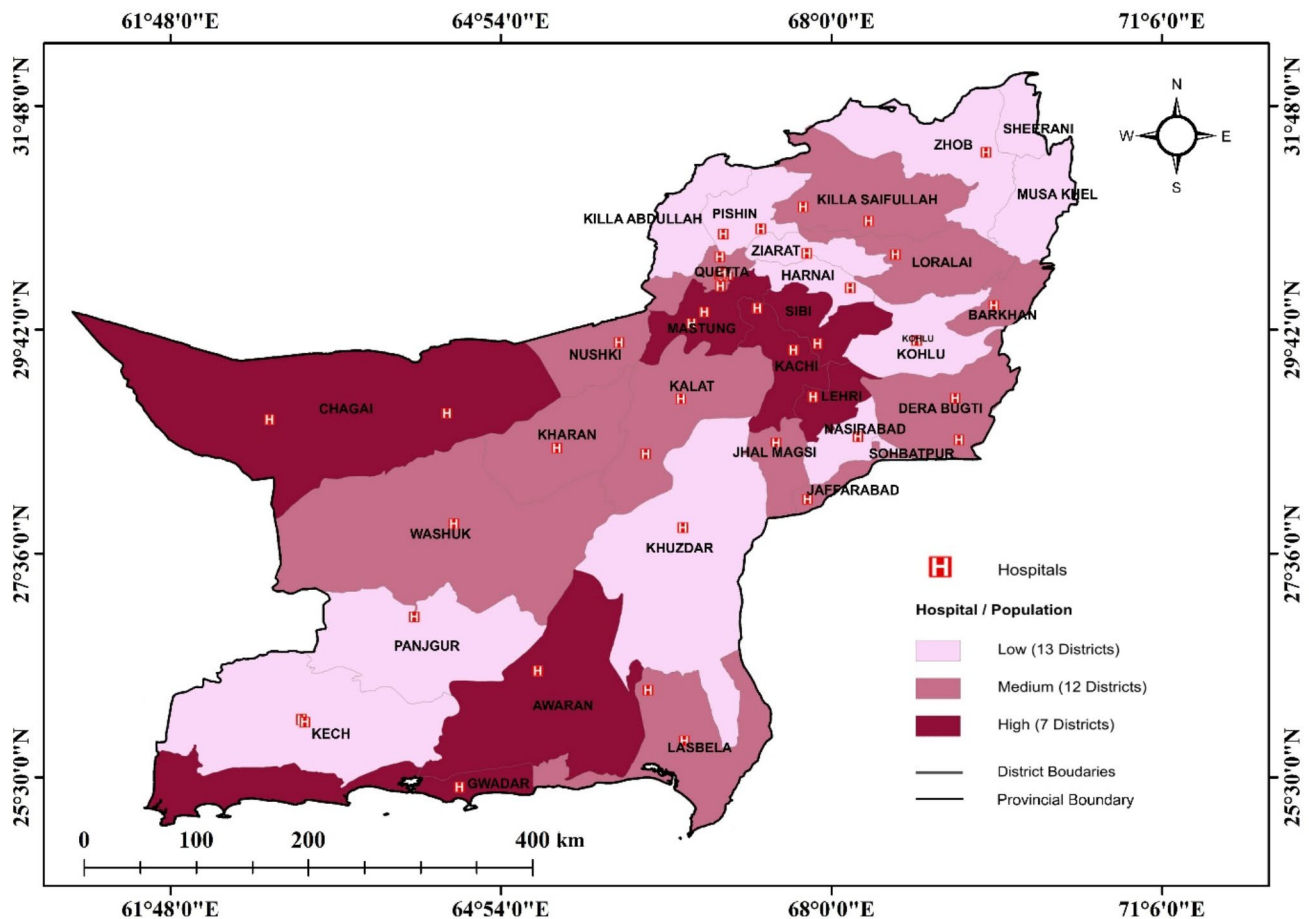


Fig. 4 The availability of hospitals at district level in Balochistan province

3.1.2 Rural Health Centers

Balochistan has a dispersed population and very low urban areas and most of the population is living in remote villages. Providing health care through RHCs is the best option for patients seeking common medical care. Figure 5 represents the spatial distribution of RHCs in Balochistan. Districts in the eastern part of the province have comparatively lesser numbers of RHCs and require a greater number of RHCs to provide essential healthcare to the population. Quetta, Kharan, and Panjgur districts also have a low number of RHCs for the population.

3.1.3 Basic health units

The dispersed population of Balochistan necessitates the establishment of primary healthcare facilities at the most basic level. Figure 6 shows that BHUs are being established at Union Council - the lowest administrative tier - levels in all districts of Balochistan to provide basic health care. However, the population burden on BHUs in all districts except Kharan, Kohlu, and Washuk is much higher. Districts

in central Balochistan, along with the northeastern part, require the establishment of more BHUs to cover the population's medical needs.

3.1.4 Civil dispensaries, mobile dispensaries, and mother & child health centers

Civil dispensaries (CDs), mobile dispensaries (MDs), and maternal and child health centres (MCHs) play a vital role in the delivery of primary health care services. The CD provides health care to the public and is headed by a physician or dispenser. MD provides health care to remote areas with no proper health care establishments by moving to those regions and catering to the health needs of isolated populations. MCH plays a vital role in the health of mothers and newborn babies and is specifically designed for mother and child health care. The CD, MD, and MCH were merged into one category, and their locations are visualized in Fig. 7.

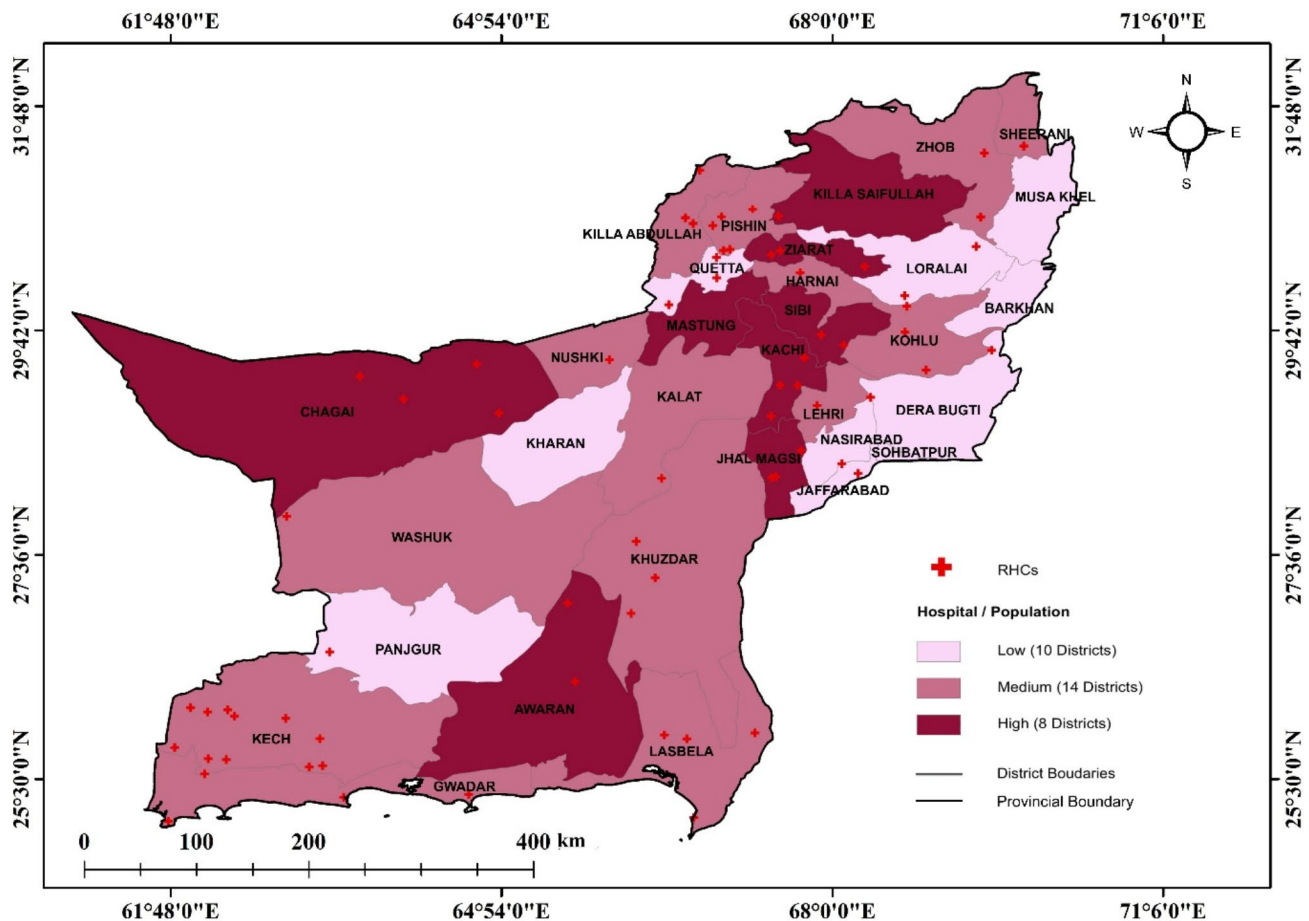


Fig. 5 The availability of RHCs at the district level in Balochistan province

3.2 Calculation of health facilities per population

The number of healthcare facilities per 10,000 population is used as an indicator (defined by WHO) to measure healthcare availability across all Balochistan districts. An equation was formulated to calculate the number of healthcare facilities per 10,000 people, as represented in (1).

$$\text{Indicator} = \frac{F \times x}{P \times x} \times 10,000 \quad (1)$$

Where F is the number of healthcare facilities, P is the population, and x is the factor required for the denominator to equal 10,000. The indicator and factor x were calculated separately for each district, and statistics are represented in Table 2.

3.3 Health facilities per 10,000 population

District-wise population distribution is measured as population/km². The density of the population/km² in a specific

district provides insights into creating new health facilities and improving present healthcare facilities in that district.

3.4 Classification of districts

Classification of districts into low, medium, and high numbers of healthcare facilities was carried out based on the division of total healthcare facilities by population and is represented in Table 3. ArcMap 10.8 automatically classifies the data based on the number of classes (i.e., low, medium, high). We classified each type of healthcare facility based on the availability and population of their respective districts.

3.5 Location summary of districts

The data gathered from national sources had missing coordinates, where the locations of health facilities were collected from Google Maps, and some remained unknown, as shown in Table 4.

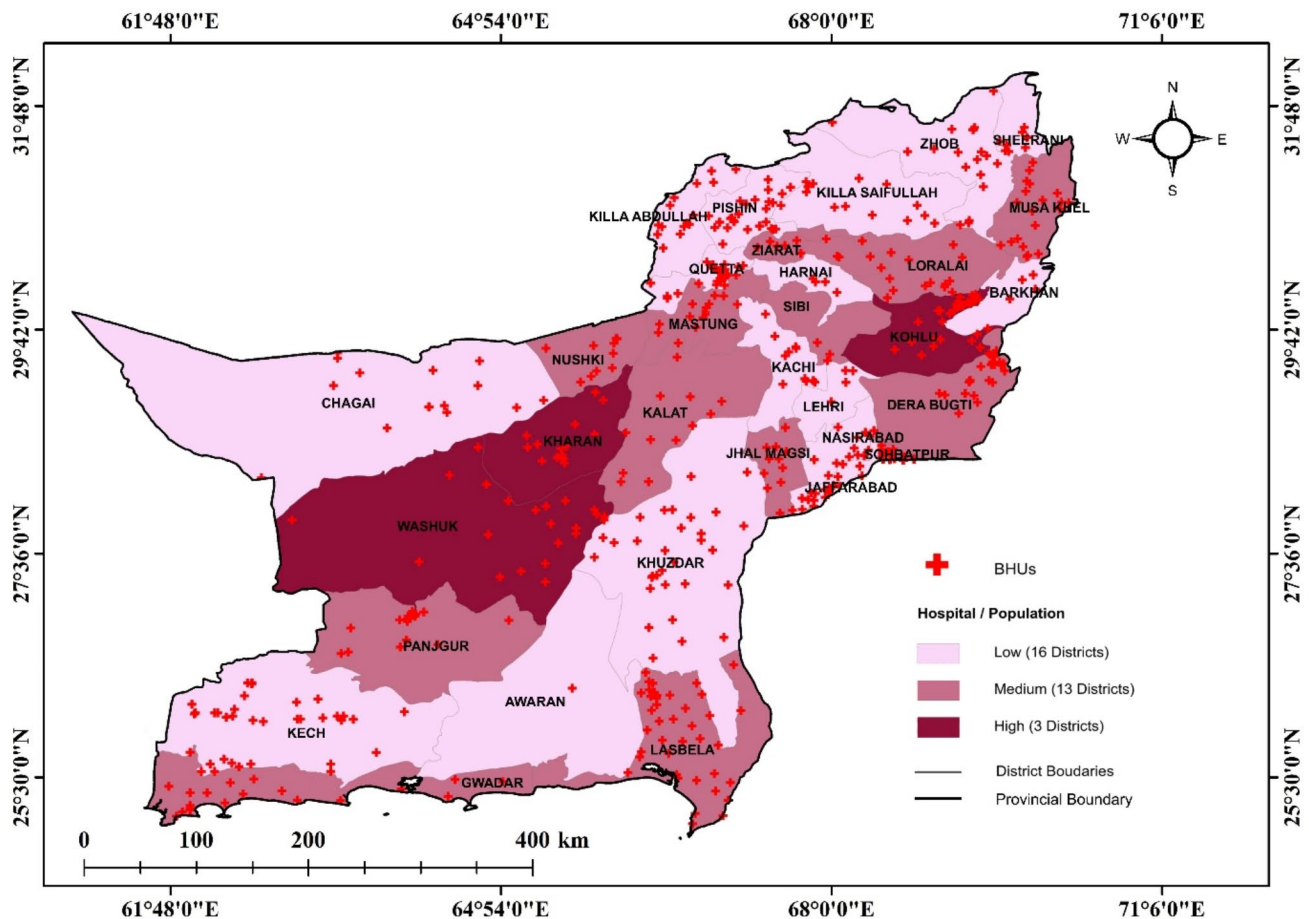


Fig. 6 The availability of BHUs at the district level in Balochistan province

4 Discussion

This study has assessed the geospatial analysis of the distribution and the availability of healthcare facilities at the district level across Balochistan province. The classification of districts into low, medium, and high is based on the availability of healthcare facilities in each district that are concerned with the population. The geospatial analysis techniques are useful for the identification of gaps and improving the efficiency of the health care system [28]. Research studies have also documented that Pakistan is lagging in providing an efficient healthcare system compared to its neighbors and low-income countries [3, 29, 30]. Thus, the current study added value to identifying gaps in the distribution of healthcare facilities in Pakistan's less-developed Balochistan province.

With regard to the availability of hospitals, the findings showed that the number of hospitals in the districts of Balochistan varies. This study categorized 13 districts as low, 12 as medium, and 7 with high availability. These findings imply that many districts need hospitals to ensure access

to health care services. Thus, more resources are required to achieve high availability at the district level across the province. The population of Quetta district is highest as it is the provincial capital. Because of the presence of hospitals, Quetta falls in the medium category. It is classified as a district with few RHCs, BHUs, CD, MD, and MCH centers per population. On the other hand, the district of Harnai has the lowest population, with a low number of hospitals and a medium number of BHUs, RHCs CD, MD, and MCH centers. Chagai district is the largest area, and the population is distributed across the entire region. Population dissemination in large districts was observed to be a major factor affecting the availability of healthcare facilities. Categorizing districts as low, medium, and high according to healthcare facilities per population represents the availability of healthcare services in the respective districts. The generation of maps assisted in analyzing the inter-district variations among the number of health facilities in Balochistan province. Assessing the availability of all types of health facilities at the district level provides a base for determining which districts require what healthcare facilities. These findings imply that the districts that are larger in area but

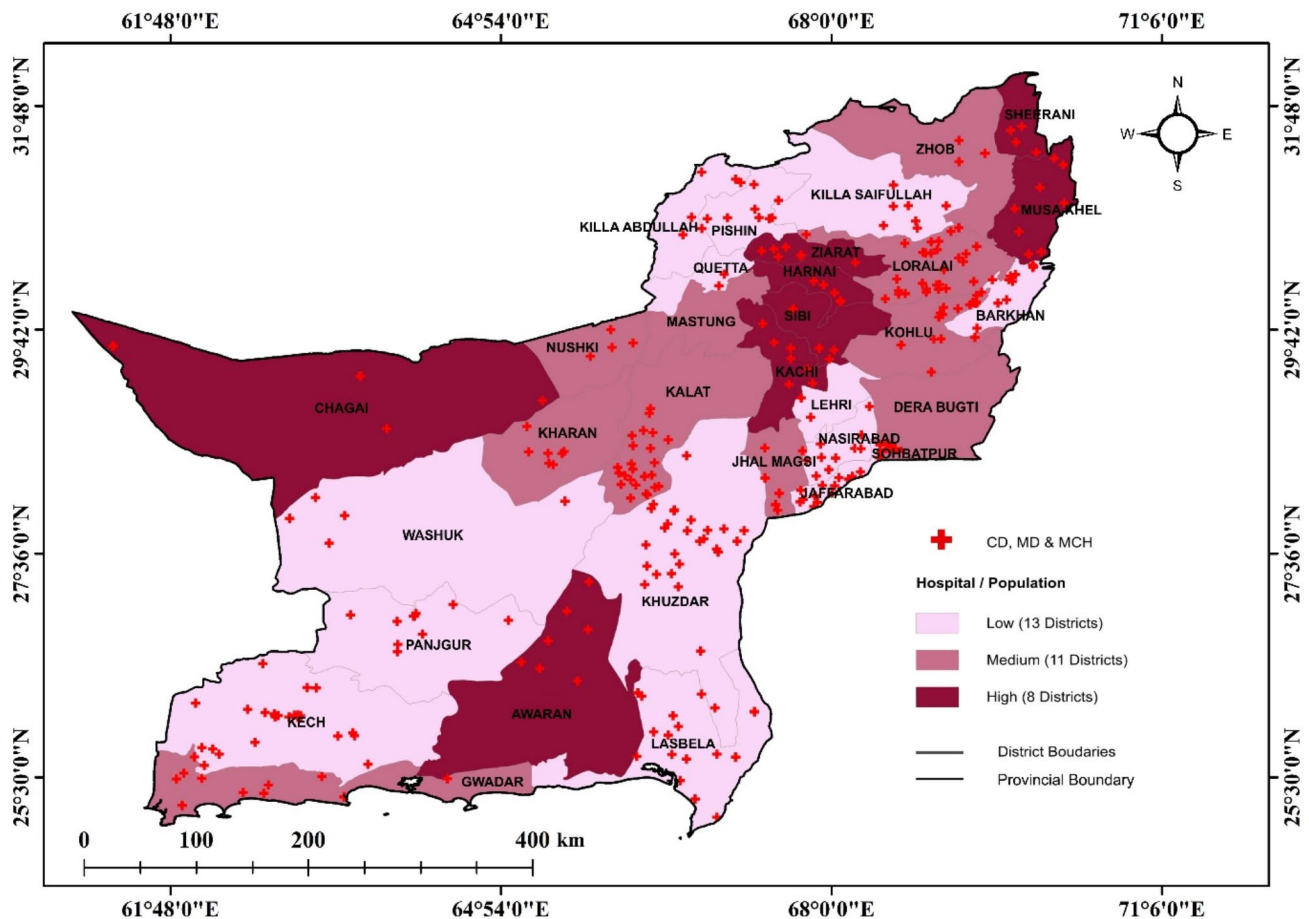


Fig. 7 The availability of CD, MD, and MCH at district level in Balochistan province

have a comparatively low population require fewer health-care establishments in strategic locations. Whereas more healthcare facilities are needed in more densely populated district areas.

The findings of the current study verify almost similar findings of previous studies by Panezai (2012) [12] who had identified inter-district variations in health care services? This study has found that flung districts of Balochistan province, such as Musa Khel, Dera Bugti, Kharan, Barkhan, Killa Abdullah, and are found to be less developed. In contrast, Quetta and Ziarat were found to be the most developed districts in terms of the delivery of health care services. The current study's findings imply that less developed districts must be prioritized for developing new healthcare facilities. Similar findings were proposed by Panezai (2012) [12] suggesting the de-concentration of health care services from the provincial capital, Quetta, and focusing on the poorly developed districts for health care infrastructure development.

Rural health centres (RHCs) are established in rural areas, i.e., tehsils of the districts, and provide PHC services to the population. These are of significant importance in the delivery of primary health care services. The current study's

findings showed that availability of RHCs was low in 10 districts, medium in 14 districts and high in 8 districts. The current study revealed that RHCs and BHUs are highly prevalent in healthcare facilities. Studies confirm that RHCs and BHUs are the facilities that are mostly located in rural areas of Pakistan [31].

Likewise, the availability of BHUs is crucial. It is the basic level of health facility for the delivery of PHC as BHUs address health care needs at the primary level. In Balochistan, the availability of BHUs per population is low in 16 districts, medium in 13 districts, and high in only 3 districts. The population burden on BHUs in all districts except Kharan, Kohlu, and Washuk is much higher. The findings imply that districts in central Balochistan, along with the northeastern part, require the establishment of more BHUs to cover the population's medical needs. Research has also reported high quality of primary health care services in the BHUs in Quetta [13]. Similarly, the availability of CD, MD, and MCH is low in 13 districts, medium in 11, and high in 8 districts.

Research studies have identified that underutilization of healthcare facilities in the Balochistan province [3, 15, 16],

Table 2 Health units per 10,000 population with people/ km² (Census 2017)

#	District	Total Health Facilities	Population	Health Facilities /10,000 pop.	Area (km ²)	People/km ²
1	Awaran	26	121,680	2.14	29,510	4.123
2	Barkhan	17	171,556	0.99	3408	50.339
3	Chagai	53	226,008	2.35	44,748	5.051
4	Dera Bugti	67	312,603	2.14	10,159	30.771
5	Gwadar	46	263,514	1.75	12,637	20.853
6	Harnai	20	97,017	2.06	2492	38.931
7	Jaffarabad	41	513,813	0.8	2445	210.148
8	Jhal Magsi	34	149,225	2.28	3615	41.279
9	Kachhi	62	237,030	2.62	5682	41.716
10	Kalat	79	412,232	1.92	8416	48.982
11	Kech	75	909,116	0.82	22,539	40.335
12	Kharan	35	156,152	2.24	14,958	10.439
13	Khuzdar	65	802,207	0.81	35,380	22.674
14	Killa Abdullah	60	757,578	0.79	4894	154.797
15	Killa Saifullah	39	342,814	1.14	6831	50.185
16	Kohlu	63	214,350	2.94	7610	28.167
17	Lasbela	66	574,292	1.15	15,153	37.9
18	Lehri	8	118,046	0.68	3285	35.935
19	Loralai	70	397,400	1.76	8018	49.563
20	Mastung	59	266,461	2.21	3308	80.550
21	Musa Khel	52	167,017	3.11	5728	29.158
22	Nasirabad	40	490,538	0.82	3387	144.830
23	Nushki	27	178,796	1.51	5797	30.843
24	Panjgur	41	316,385	1.29	16,891	18.731
25	Pishin	56	736,481	0.76	6218	118.443
26	Quetta	69	2,275,699	0.3	3447	660.197
27	Sherani	33	153,116	2.16	4310	35.526
28	Sibi	36	135,572	2.65	7121	19.038
29	Sohbatpur	42	200,538	2.09	802	250.047
30	Washuk	38	176,206	2.16	33,093	5.325
31	Zhob	46	310,544	1.48	15,987	19.425
32	Ziarat	46	160,422	2.87	3301	48.598

Note: Total health facilities=Hospital, RHC, BHU, CD, MD, and MCH

Sources: [26, 27]

Table 3 The range for classification of a district into low, medium, and high categories

Health Facility Type	Range		
	Low	Medium	High
Hospitals	0.0000–0.0000032	0.0000033–0.0000067	0.0000068–0.0000088
RHCs	0.000000–0.00000632	0.00000633–0.0000139	0.000014–0.000022
BHUs	0.0000175–0.0000721	0.0000722–0.0001197	0.0001198–0.0001866
CD, MD & MCH	0.0000071–0.0000437	0.0000438–0.00011	0.00012–0.000185

in Punjab province [32], Sindh and Khyber Pakhtunkhwa province [33]. The reasons for underutilization include the unavailability of doctors, poor infrastructure facilities at BHUs, lack of obstetric care, inconvenient location of BHUs, and absenteeism of BHU staff. These findings imply that the mere availability of healthcare facilities is not enough. Several other service quality factors must be ensured at healthcare facilities, such as ease of access, availability of doctors, provision of diagnostic services, presence of trained supporting staff, and length of sufficient service

hours. Thus, besides establishing healthcare facilities in less-developed districts, policymakers and healthcare managers must also priorities the quality of healthcare [34].

This study has a few limitations. Districts in Balochistan have increased from 32 to 35. Due to the non-availability of the latest population, district boundaries, and healthcare facilities data, the three districts created after the 2017 census were not included separately. Therefore, we must interpret the study's results cautiously. Secondly, there is a large number of registered and non-registered Afghan

Table 4 Location summary of healthcare facilities at the district level in Balochistan

Districts	Hospitals			RHCs			BHUs			CD, MD & MCH		
	Total	Avail.	Miss.	Total	Avail.	Miss.	Total	Avail.	Miss.	Total	Avail.	Miss.
Awaran	1	1	0	2	2	0	7	1	6	16	7	9
Barkhan	1	1	0	0	0	0	10	7	3	13	11	2
Chagai	2	2	0	4	4	0	15	15	0	8	7	1
Dera Bugti	2	2	0	1	1	0	34	28	6	24	0	24
Gwadar	2	2	0	3	3	0	23	20	3	16	0	16
Harnai	1	1	0	1	1	0	7	5	2	8	5	3
Jaffarabad	3	3	0	1	1	0	19	19	0	20	19	1
Jhal Magsi	1	1	0	3	3	0	14	12	2	12	12	0
Kachhi	2	2	0	5	4	1	12	12	0	16	7	9
Kalat	2	2	0	3	1	2	43	14	29	40	25	15
Kech	2	2	0	12	12	0	39	38	1	40	23	17
Kharan	1	1	0	0	0	0	22	15	7	13	7	6
Khuzdar	1	1	0	6	3	3	45	31	14	30	28	2
Killa Abdullah	1	1	0	6	3	3	38	15	23	10	7	3
Killa Saifullah	2	2	0	5	1	4	17	14	3	15	7	8
Kohlu	1	1	0	3	3	0	40	35	5	30	17	13
Lasbela	3	3	0	4	4	0	44	41	3	26	24	2
Lehri	1	1	0	1	1	0	4	4	0	2	2	0
Loralai	2	2	0	2	2	0	42	15	27	47	30	17
Mastung	2	2	0	4	0	4	24	14	10	6	0	6
Musa Khail	1	1	0	1	0	1	20	20	0	14	10	4
Nasirabad	1	1	0	2	2	0	18	14	4	10	10	0
Nushki	1	1	0	2	1	1	14	10	4	18	5	13
Panjgur	1	1	0	2	1	1	29	20	9	13	10	3
Pishin	2	2	0	9	6	3	34	31	3	11	8	3
Quetta	9	9	0	4	3	1	40	36	4	6	4	2
Sherani	2	2	0	5	1	4	17	14	3	15	7	8
Sibi	1	1	0	3	3	0	40	35	5	30	17	13
Sohbatpur	3	3	0	4	4	0	44	41	3	26	24	2
Washuk	1	1	0	1	1	0	4	4	0	2	2	0
Zhob	2	2	0	2	2	0	42	15	27	47	30	17
Ziarat	2	2	0	4	0	4	24	14	10	6	0	6
Total	52	52	0	100	74	26	765	566	199	540	344	196

Note. Hospitals = Tehsil Headquarters (DHQ) Hospitals and District Headquarters (DHQ) Hospitals and teaching hospitals

refugees in Balochistan. Their official statistics are not yet known. These Afghan refugees access the health care facilities across the province and have caused a burden on the current health care system of the province. The absence of refugees' data in this study could have influenced the findings if we had included their population data. Thirdly, this study has covered only public healthcare facilities. The findings would have differed if the analysis had included private healthcare facilities.

5 Conclusion

This study aimed to assess the geospatial analysis of the distribution and the availability of healthcare facilities at the district level across Balochistan province. Considering availability, healthcare facilities across Balochistan's district levels were categorized into 3 categories: low, medium, and high. Districts with vast areas with sparse yet lower populations are prone to lower levels of healthcare facility availability. This study found distinct inter-district variations in the availability of healthcare facilities. The remote districts were found to be less developed compared to the districts near the provincial capital, Quetta. The research findings show that Balochistan is lagging in providing health care services to its people. Considering the geospatial analysis, the research findings highlight the gaps and shortcomings in the availability of healthcare services across the province. Therefore, districts where the availability of healthcare facilities was categorized as low require the establishment of new healthcare infrastructure. The current study's findings are useful to the government and non-government organizations planning to invest in improving the health care delivery system in Balochistan province. Developing new and enhanced healthcare infrastructure will strengthen the efforts to build a healthier community. This study attempted to assess a quantitative analysis of healthcare facilities. Future research may be conducted on the quality factors of the healthcare delivery system of the Balochistan province.

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Data availability Data and materials for the present study will be available upon request.

Declarations

Conflict of interest The authors declare no conflict of interest.

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